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Suspicious Activity Detection in Exam Hall using Deep Learning

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Abstract: *Cheating in academic examinations compromises the integrity and fairness of educational assessments. Manual invigilation alone is often insufficient, especially in large-scale exams, where continuous monitoring is impractical. This study proposes a real-time, deep learning-based system for detecting suspicious activity in examination halls using surveillance footage. By leveraging Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, the system analyzes student behavior across video frames to flag anomalies such as unusual head movements, note passing, or collaboration. The solution enhances accuracy through temporal behavior analysis and operates non-intrusively using standard CCTV setups. Designed for scalability and minimal human oversight, the system aims to elevate examination security while maintaining student privacy.*

Keywords: *Exam Surveillance, Deep Learning, CNN-LSTM, Suspicious Activity Detection, Computer Vision, Real- Time Monitoring, Academic Integrity.*

I. INTRODUCTION

With cheating during academic examinations becoming increasingly sophisticated, maintaining assessment integrity is a pressing concern for educational institutions. Studies have shown that traditional manual invigilation is often inadequate in large exam settings, where human error, limited visibility, and the inability to monitor every candidate simultaneously lead to frequent lapses in oversight. Existing rule-based or sensor-dependent approaches for suspicious activity detection are often expensive, intrusive, or unsuitable for dynamic exam environments.

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Recent advancements in deep learning and computer vision have enabled real-time video analysis for behavior monitoring. Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks are particularly effective at extracting spatial features and learning temporal behavioral patterns from surveillance footage. These models provide an edge in distinguishing normal student behavior from suspicious movements, such as unusual head orientation, contact with peers, or passing objects.

This research proposes a deep learning-based real-time surveillance system that processes live CCTV video streams to detect potential cheating behaviors during exams. The system combines CNNs for spatial detection and LSTMs for temporal analysis, enabling it to recognize behavioral anomalies like frequent glancing, peer interactions, or body orientation changes. By providing instant alerts and detailed incident logging, the system minimizes manual intervention and enhances the fairness and transparency of the examination process. Future improvements include adapting to varied lighting conditions and integrating multimodal behavior cues for increased robustness.

II. SCHEDULING AND ALGORITHMIC APPROACHES

In the proposed cheating detection system, algorithmic design and scheduling are crucial to ensure real-time detection, accuracy, and efficient utilization of processing resources during examinations. The system is structured as a pipeline of sequential yet highly optimized modules, each responsible for specific tasks such as frame extraction, preprocessing, student detection, and suspicious activity classification.

Upon receiving the live video feed, the system first segments it into frames and performs preprocessing operations. These frames are then analyzed by Convolutional Neural Networks (CNNs) to extract spatial features—such as student posture, head orientation, and interactions with nearby objects. The output of this stage is forwarded to Long Short-Term Memory (LSTM) networks, which evaluate behavioral patterns over time to distinguish between regular and suspicious activities.

To maintain real-time performance, frame processing is scheduled in fixed intervals, with behavior checkpoints analyzed every few seconds. The system's pipeline ensures parallel processing where feasible—for example, head motion detection, contact tracking, and activity recognition modules operate concurrently to minimize latency. If the model detects actions like frequent head-turning, peer interaction, or hand movement that exceeds learned behavioral thresholds, it raises an alert immediately.

Furthermore, the model employs lightweight architecture and is designed to run efficiently even on modest hardware configurations typical in examination centers. This ensures that alerts are generated with minimal delay, allowing invigilators to respond promptly.

Overall, the integration of disciplined scheduling and intelligent algorithmic layers enables the system to function reliably in dynamic exam hall environments, improving surveillance quality without adding computational overhead.

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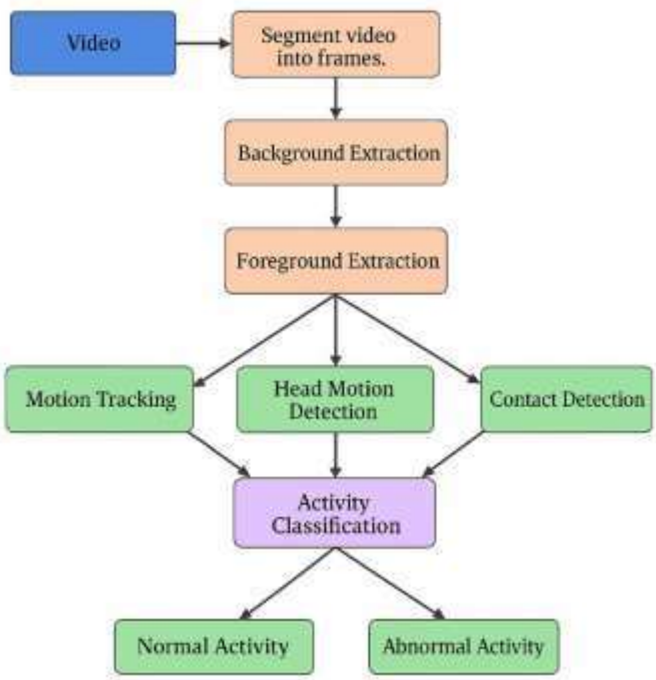
III. LITERATURE REVIEW

Sr. No	Title	Year	Objective	Methodology	Advantages	Future Scope
1	Suspicious Activity Detection using LSTM	2024	Identify cheating by analyzing behavior patterns in exam videos	LSTM-based behavior detection: video preprocessing, feature extraction, suspicious pattern check	Achieves 90–95% accuracy, real-time detection	Enhance generalization across varied exam environments
2	Aptitude Assessment System for Skill Evaluation	2024	Provide adaptive aptitude tests and evaluate skill proficiency	Adaptive Testing Algorithms	Personalized evaluations, real-time tracking	Integrate emotional response tracking with AI
3	Detecting Suspicious Activities in Surveillance using Deep Learning	2023	Classify head motion and track abnormal behavior in students	Head motion detection, motion tracking, deep learning-based activity classification	84–85% accuracy, robust motion tracking	Combine with multimodal (audio, biometrics) inputs
4	Toward Trustworthy Human Suspicious Activity Detection	2023	Improve detection trust and accuracy in surveillance	Hybrid CNN/RNN architecture, robust training with deep features	85–89% accuracy, high adaptability	Expand use to real-time mobile edge devices
5	Automatic Cheating Detection in Exam Hall	2023	Detect exam cheating using vision-based automation	YOLOv3 + Residual Networks on CCTV footage	90–92% accuracy, high-speed analysis	Implement cloud alert system for large campus use

IV. PROPOSED SYSTEM DESIGN

System Architecture

- Input: Real-time video feed from surveillance CCTV cameras in exam halls.
- Frame Segmentation: The video stream is broken into individual image frames for analysis.
- Person and Landmark Detection: Using deep learning (CNN/YOLO), the system identifies
students' heads, facial orientation, and body posture.
- Feature Extraction: The system extracts key behavioral features such as head direction, motion intensity, hand movement, and proximity to nearby students.
- Suspicious Behavior Classification: A trained deep neural network evaluates these features using temporal analysis to classify behaviors as normal or suspicious.
- Alert System: If suspicious behavior exceeds a predefined threshold (e.g., consistent head turning, repeated movement), the system generates a real-time alert for invigilators.
- Data Logging**:** All flagged incidents are logged with timestamps, frame snapshots, and behavior metadata for post-exam review and evidence.



V. SOFTWARE REQUIREMENTS SPECIFICATION

Functional Requirements

- Detect student presence and head orientation using deep learning models.
- Monitor movements such as head turns, hand gestures, and leaning behaviors.
- Classify activity as normal or suspicious using trained CNN models.
- Generate alerts and logs when suspicious behavior is detected.
- Store incident data (frames, time, student ID) for later review.

Non-Functional Requirements

- Real-time video processing with minimal delay.
- High accuracy in detecting cheating-related behavior.
- Scalable for use across multiple exam halls simultaneously.
- Low hardware resource consumption.
- User-friendly GUI for invigilators and post-exam analysis.

System Requirements

- **Hardware:** CCTV camera or high-resolution webcam, minimum Intel i5 processor, 8 GB RAM, GPU (optional for faster processing)
- **Software:** Python 3.x, OpenCV, TensorFlow or PyTorch, SQLite (for logs), Pre-trained YOLO/CNN models, DB Browser (optional)

VI. COMPARISON WITH EXISTING SYSTEMS

Sr No.	Feature	Implemented System	Existing Manual Systems	Basic Automated Systems
1	Real-time detection	Yes	Limited	Partial (based on periodic checks)
2	Suspicious movement detection	Yes (CNN/LSTM-based pattern recognition)	No (observer dependent)	Partial (motion sensors)
3	Face and posture monitoring	Yes (facial & body landmark detection)	Limited	Rare
4	Instant alert generation	Yes (threshold-based triggers with logging)	No	Limited (basic alert after review)
5	Adaptability to lighting conditions	High (with preprocessing and filters)	Poor	Moderate
6	Intrusiveness	Non-intrusive (uses existing CCTV setup)	High (requires active invigilation)	Moderate (may require specialized devices)
7	Data logging for evidence	Yes (image snapshots + metadata)	No	Limited (basic timestamps only)
8	Scalability and upgradability	High (modular system for multiple halls)	Very Low	Low
9	Computational efficiency	High (optimized models with batch frame processing)	Not Applicable	Moderate
10	Overall accuracy	High (85–92% depending on model used)	Low	Moderate (depends on configuration)

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VII. GUI

Figure 2: Home Page.

Figure 3: Main Page.

Figure 4:Upload video.

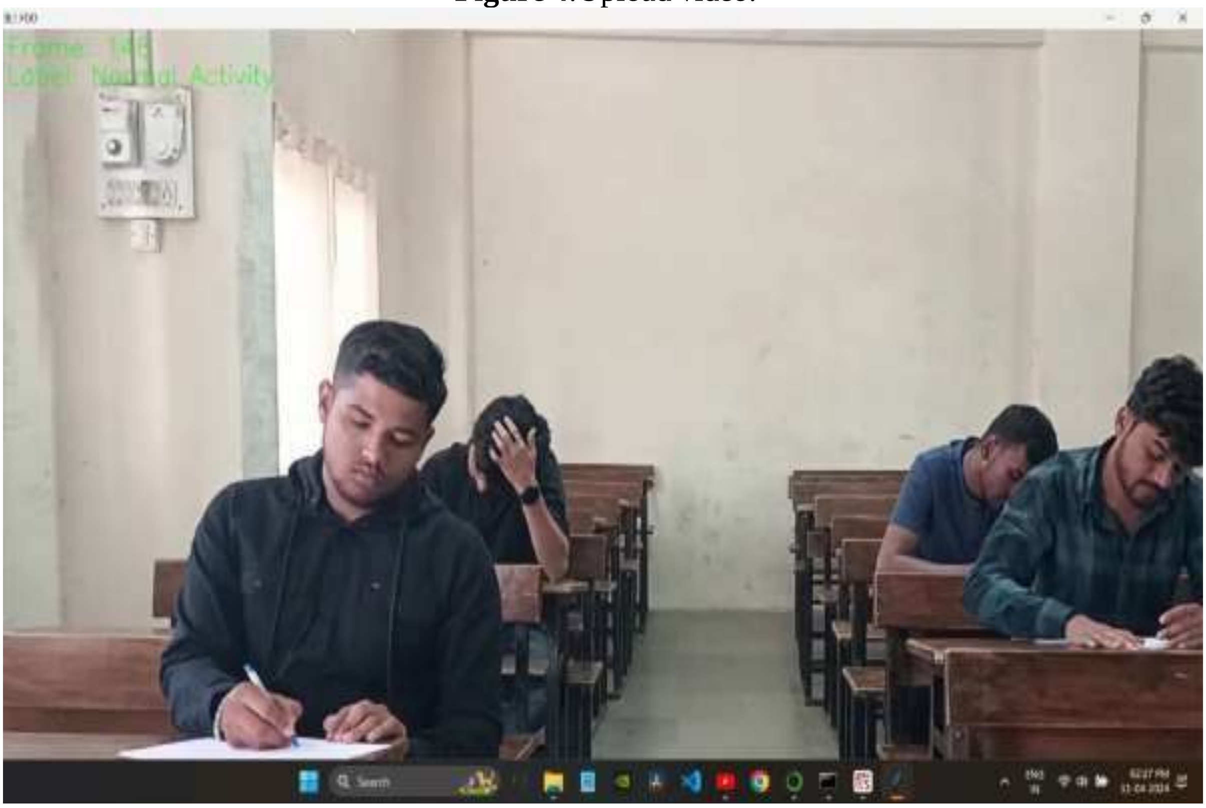

Figure 5: Normal Activity.

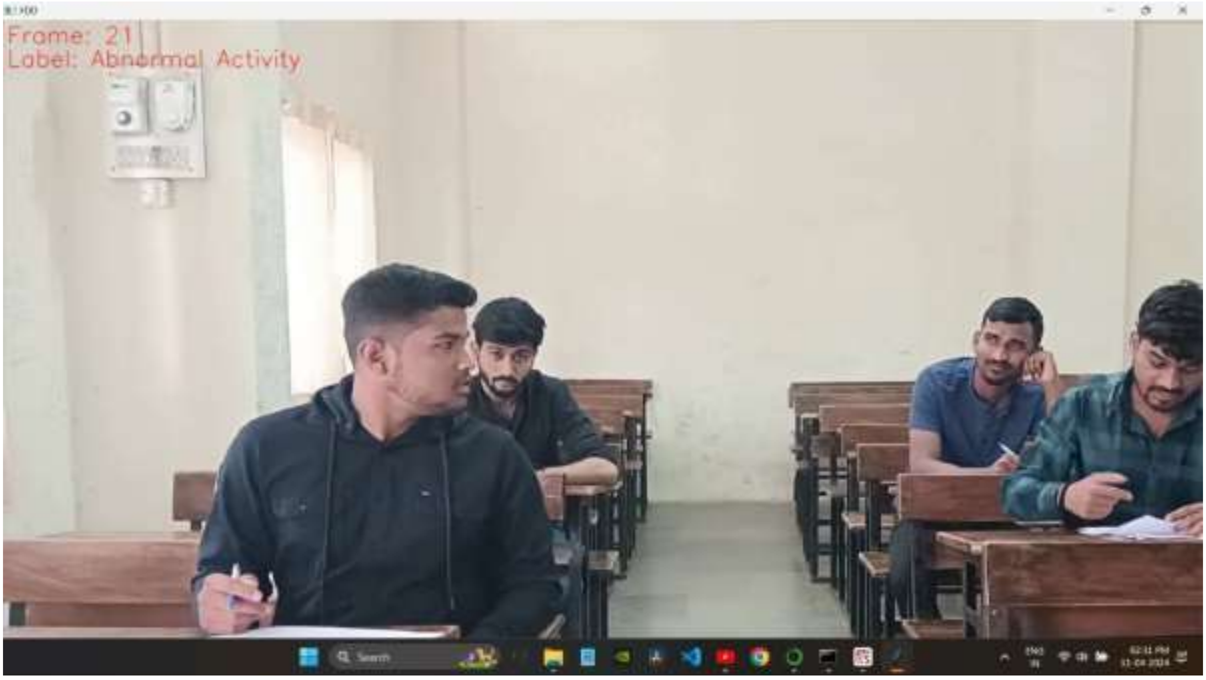

Figure 6: Abnormal Activity

VIII. Conclusion and Future Work

This study presents a real-time cheating detection system using deep learning and computer vision to monitor suspicious behavior during examinations. By utilizing advanced models such as CNNs and LSTMs, the system can accurately detect actions like head turning, leaning, and hand movements— common indicators of academic dishonesty. Through efficient video preprocessing and pattern recognition techniques, the system ensures timely alerts and incident logging with minimal latency.

Compared to traditional manual invigilation and basic rule-based automated systems, the proposed solution demonstrates substantial improvements in behavior recognition accuracy, response time, and scalability. Its non-intrusive nature and compatibility with existing CCTV infrastructure make it practical for deployment across multiple educational environments.

Future enhancements may include the integration of multimodal data sources like audio cues or biometric signals, behavior profiling using reinforcement learning, and real-time dashboard analytics for exam administrators. Overall, the system marks a significant step toward intelligent, technology- driven exam supervision that promotes academic fairness and integrity.

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